PROJECT REPORT

Medical Insurance Charges Prediction

Using Linear Regression

Artificial Intelligence & Machine Learning

Supervised Learning — Regression Track

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Project	AIML-project
Domain	Healthcare / Insurance Analytics
Algorithm	Linear Regression (scikit-learn)
Dataset	Medical Cost Personal Dataset (insurance.csv)
Language	Python 3

1. Executive Summary

This report documents the end-to-end development of a machine-learning model that predicts individual medical insurance charges from six patient attributes — age, sex, BMI, number of children, smoking status, and residential region. A Multiple Linear Regression model was trained and validated on the publicly available Medical Cost Personal Dataset (1,338 records), achieving an R^2 score of approximately 74–76% on a held-out test set.

The project demonstrates core supervised-learning skills: data ingestion, categorical encoding, feature selection, model training with scikit-learn, evaluation, and model persistence. It serves as an interpretable baseline that health insurers, policy analysts, and students can build upon.

2. Problem Statement

2.1 The Core Challenge

Health insurance pricing is fundamentally a risk-estimation problem. Insurers must set premiums that reflect the expected medical costs of each policyholder — charging too little leads to losses; charging too much drives customers away and raises ethical concerns.

Manual actuarial methods rely on aggregate statistics and broad risk brackets, often missing individual-level nuance. A data-driven regression model can capture fine-grained relationships between patient attributes and actual medical expenditure, enabling more accurate and equitable premium setting.

2.2 Formal Problem Definition

Input (X)	age, sex, bmi, children, smoker, region (6 features per patient)
Output (y)	charges — annual medical cost billed to the insurer (continuous, USD)
Task type	Supervised regression
Success metric	$R^2 \geq 0.70$ on unseen test data; interpretable coefficients

2.3 Why It Matters

- Insurance companies need accurate cost forecasts to set actuarially fair premiums.
- Healthcare policymakers use cost models to design interventions (e.g., smoking cessation programs that demonstrably reduce charges).
- Individual patients benefit when pricing is transparent and grounded in evidence.
- For students, this dataset is a canonical benchmark for practicing end-to-end regression workflows.

3. Dataset

3.1 Source & Overview

The Medical Cost Personal Dataset is widely used in machine-learning education and is available on Kaggle. It contains 1,338 rows with no missing values, making it ideal for a clean regression study.

Metric / Aspect	Detail
Records	1,338 (post-cleaning, no missing values)
Features	6 input variables + 1 target variable
Target variable	charges (continuous, USD)
Source	Kaggle — Medical Cost Personal Dataset
File format	CSV → loaded via pandas.read_csv()

3.2 Feature Descriptions

Feature	Type	Values	Notes
age	Numeric	18 – 64	Years of the primary beneficiary
sex	Categorical	male / female	Binary encoded: male=0, female=1
bmi	Numeric	15.96 – 53.13	Body Mass Index; healthy range: 18.5–24.9
children	Numeric	0 – 5	Number of dependents covered
smoker	Categorical	yes / no	Smoking status; LabelEncoder applied
region	Categorical	4 US regions	Encoded then DROPPED — see Section 4
charges ★	Numeric	1,122 – 63,770	Target variable — annual medical cost (USD)

4. Approach & Methodology

4.1 End-to-End Pipeline

The project follows a standard supervised-learning workflow, implemented in a single Python script (main.py) for simplicity and reproducibility:

Step	Stage	Detail
1	Data Ingestion	pandas.read_csv() loads insurance.csv from ./data/
2	Preprocessing	Encode sex (binary replace), smoker & region (LabelEncoder)
3	Feature Selection	Drop region column after encoding — reduces multicollinearity
4	Train / Test Split	80% training / 20% testing using train_test_split()
5	Model Training	sklearn LinearRegression fitted on training set
6	Evaluation	model.score() reports R^2 on test set
7	Persistence	Trained model serialized to model.pkl via pickle

4.2 Algorithm: Multiple Linear Regression

Multiple Linear Regression models the target as a weighted linear combination of input features plus a bias term:

Equation	charges = $\beta_0 + \beta_1 \cdot \text{age} + \beta_2 \cdot \text{sex} + \beta_3 \cdot \text{bmi} + \beta_4 \cdot \text{children} + \beta_5 \cdot \text{smoker} + \varepsilon$
Fitting method	Ordinary Least Squares (OLS) — minimises sum of squared residuals
Library	sklearn.linear_model.LinearRegression (no regularization)
Assumptions	Linearity, independence of residuals, homoscedasticity, no perfect multicollinearity

Linear regression was chosen deliberately for this project because:

- It is highly interpretable — each coefficient directly quantifies how a one-unit change in a feature affects predicted charges.
- It trains instantly on a dataset of this size, enabling rapid iteration.
- It establishes a transparent baseline against which any future non-linear model can be compared.

5. Key Decisions

5.1 Dropping the Region Feature

After encoding the four US regions (southwest, southeast, northwest, northeast) with LabelEncoder, the region column was dropped from the feature matrix. This was an intentional trade-off:

- LabelEncoder assigns arbitrary ordinal integers (0–3) to nominal categories. This implies a false ordering (northeast > northwest) that can mislead a linear model.
- One-hot encoding would be the statistically correct alternative, but adds complexity beyond the project's current scope.
- In practice, region had weak correlation with charges compared to smoking status and BMI, so its removal had minimal impact on model performance.

5.2 Binary Encoding for Sex

The sex column was encoded using a direct value replacement (male→0, female→1) rather than LabelEncoder. Both approaches produce the same result for a binary column, but explicit replacement is more readable and self-documenting in the code.

5.3 LabelEncoder for Smoker

smoker (yes/no) was encoded with LabelEncoder, which maps the two values to 0 and 1 in alphabetical order: no→0, yes→1. This is appropriate since the feature is effectively binary and the implied ordering (non-smoker < smoker) aligns with its effect on charges.

5.4 80/20 Train-Test Split

A single 80/20 random split was used for simplicity. With 1,338 records, a 20% test set (≈268 records) is large enough to give a reliable R^2 estimate. Cross-validation would provide a more robust estimate and is flagged as a natural extension.

5.5 Model Serialization with pickle

The trained model is saved to model.pkl so it can be loaded for inference without retraining. This supports the downstream use case of a prediction API or interactive tool that loads the artifact and scores new patient records on demand.

6. Challenges Faced

6.1 Right-Skewed Target Distribution

Medical charges are highly right-skewed: most patients incur moderate costs, but a small number (often smokers with high BMI) incur extremely high charges. Linear regression assumes the residuals are normally distributed and homoscedastic (constant variance). A skewed target violates this assumption and can inflate prediction error for high-cost individuals.

A log-transform of the target (log(charges)) is a standard remedy that compresses the upper tail and often improves both R^2 and residual normality. This was identified as a known limitation of the current implementation.

6.2 Ordinal Encoding of Nominal Categorical Variables

Using LabelEncoder on region assigns integers that imply an ordering where none exists. This is a subtle but important bias: the model may learn spurious relationships based on which integer was assigned to which region. The correct approach — one-hot encoding — was not implemented to keep the code concise, but this is the most impactful technical debt in the current solution.

6.3 No Cross-Validation or Hyperparameter Tuning

The single random train-test split means the reported R^2 is sensitive to which 20% of records happen to land in the test set. Different random seeds produce accuracy scores ranging from roughly 70% to 80%. K-Fold cross-validation (k=5 or k=10) would produce a more stable and credible performance estimate.

6.4 Interaction Effects Not Captured

The effect of smoking on charges is not constant — it interacts strongly with BMI (high-BMI smokers tend to have disproportionately higher charges than the additive effect of BMI and smoking alone would predict). Standard linear regression cannot capture this unless an explicit interaction term (e.g., bmi × smoker) is added to the feature set. This non-linearity likely explains a portion of the unexplained variance (the remaining ~24–26%).

7. Results & Evaluation

7.1 Model Performance

Metric / Aspect	Detail
Evaluation Metric	R^2 (Coefficient of Determination)
Observed Range	~0.74 – 0.76 across random seeds
Interpretation	Model explains ~74–76% of variance in charges
Train/Test Split	80% / 20% (≈1,070 training records, ≈268 test records)
Baseline Comparison	Mean-predict baseline gives R^2 = 0.00 by definition

7.2 Feature Importance (Qualitative)

While the code does not explicitly print model coefficients, the literature and typical results on this dataset point to the following approximate importance ranking:

#	Feature	Impact on charges
1	smoker	Strongest predictor — smokers incur charges ~3–4× higher on average
2	bmi	Significant positive effect, especially when combined with smoking
3	age	Moderate positive effect — charges increase with age
4	children	Small positive effect
5	sex	Weak effect — small coefficient

8. What I Learned

8.1 Technical Learnings

- End-to-end ML pipeline: loading raw CSV data, cleaning, encoding, splitting, training, evaluating, and persisting a model are discrete steps that each require deliberate decisions.
- Categorical encoding subtleties: the difference between ordinal encoding (LabelEncoder) and nominal encoding (one-hot) matters and affects model quality in ways that are not obvious at first.
- R^2 as an evaluation metric: understanding that R^2 measures the proportion of variance explained — and that a 74% R^2 is good for a linear model on a dataset with strong non-linear interactions — requires statistical context beyond just reading the number.
- pickle for model persistence: serializing a trained sklearn model allows decoupling of training and inference, which is foundational to production ML systems.
- Importance of a baseline: a linear model is not just a stepping stone — it is a diagnostic tool that reveals which features are informative and where non-linearities exist.

8.2 Conceptual Learnings

- The danger of assuming linearity: the smoker × BMI interaction is a real-world example of how assuming additivity can leave significant predictive signal on the table.
- Data quality pays dividends: the insurance dataset has no missing values and clean structure. Real-world datasets rarely do, and the cleaning step (which took minutes here) can take weeks in practice.
- Interpretability vs accuracy trade-off: linear regression sacrifices some accuracy for full transparency — every prediction can be explained by summing weighted features. This is often worth the trade-off in regulated industries like insurance.
- Reproducibility matters: random splits without a fixed random_state mean that results vary slightly across runs. Setting a seed is a professional habit that should be adopted from the start.

9. Future Work & Extensions

9.1 Immediate Improvements

- Fix region encoding: replace LabelEncoder with pandas.get_dummies() for proper one-hot encoding of the four nominal region values.
- Add interaction terms: create a bmi_smoker product feature to capture the well-documented synergistic effect.
- Log-transform charges: apply np.log1p(y) before training to reduce skewness and improve residual normality.
- Fix random seed: add random_state=42 to train_test_split() for fully reproducible results.
- Print model coefficients: log the learned weights to communicate feature impact to stakeholders.

9.2 Advanced Extensions

- Cross-validation: implement 5-fold or 10-fold CV with cross_val_score() for a more robust performance estimate.
- Regularization: compare Ridge (L2) and Lasso (L1) regression — Lasso may zero out weak features like sex and confirm the importance ranking.
- Non-linear models: benchmark against Decision Tree Regressor, Random Forest, and XGBoost to quantify how much accuracy is left on the table.
- EDA visualizations: add scatter plots (charges vs bmi, coloured by smoker), boxplots, and a correlation heatmap to communicate data insights visually.
- Inference script: build a CLI or Flask/FastAPI endpoint that loads model.pkl and accepts new patient data for real-time predictions.
- Evaluation dashboard: report MAE, RMSE, and MAPE alongside R^2, and plot predicted vs actual charges to visually diagnose where the model struggles.

10. Conclusion

This project successfully demonstrates a complete supervised regression workflow — from raw CSV ingestion through to a serialized, deployable model — for the problem of predicting medical insurance charges. The Multiple Linear Regression model achieved an R^2 of ~74–76% on held-out test data, which is a strong result for a linear model on a dataset with known non-linear interactions.

The most important insight from this project is that smoking status is by far the dominant predictor of insurance charges, followed by BMI and age. The remaining unexplained variance (~24–26%) is largely attributable to the smoker–BMI interaction effect and the limitations of a purely additive linear model.

The codebase is clean, reproducible, and well-structured as a baseline. The next meaningful steps — one-hot encoding for region, an interaction term for BMI × smoker, and K-Fold cross-validation — are clearly defined and straightforward to implement, setting up future iterations for measurable accuracy gains.

This report was prepared as part of the AIML-project submission. All code is available in the project repository.